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A New Method for Conserving Energy in Gas Production Plants by Wellstream Expansion and Waste Energy Extraction

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Abstract

This paper evaluates the integration of a turboexpander–generator (TXG) system into offshore gas production facilities as an alternative to conventional topside choke valves (CCV). Using process simulations over a 23-year production profile, the study quantifies the system's impact on platform power balance, emissions reduction, condensate recovery, and economic performance. Results indicate that replacing conventional throttling with near-isentropic expansion can generate a significant share of the platforms power requirement while reducing fuel gas consumption, leading to an estimated 222,000-tonne reduction in CO_2 emissions over the field life. The substantial cooling effect from expansion also increased condensate recovery by approximately 309,000 Sm^3.

Beyond the plateau phase, the system demonstrated continued value during production decline by rerouting recycled gas streams from compressors to recover otherwise lost energy. Additionally, the configuration can be adapted to accommodate new tie-in fields, enabling sustained energy recovery and process optimization after the original field transitions off plateau production. An upstream heater was incorporated to control downstream temperatures, enabling the use of waste heat from gas turbines to increase gas enthalpy prior to expansion, effectively converting otherwise wasted heat into electrical energy. The study also proposes avoiding re-bundling of export or re-injection compressors during off-design operation by recycling gas back to the TXG system to produce power and exploit waste energy sources as decline progresses.

This paper also assesses the increased range of production and energy-efficiency optimization strategies enabled by the TXG system.

Net present value (NPV) analyses indicated robust economic performance, with internal rates of return between 23% and 63%, The P50 and P10 cases were profitable without carbon tax incentives. These findings demonstrate that TXG systems can extend field life, enhance process efficiency, and reduce emissions while maintaining operational flexibility, supporting offshore decarbonization strategies.

Introduction

Gas production platforms typically rely on gas turbines for power generation, while wellstream pressure reduction is performed through (isenthalpic) choke valves. This approach dissipates the available energy stored in the reservoir providing minimal utility to the operator. Offshore facilities are commonly powered by simple-cycle gas turbines. These machines have relatively low thermal efficiency and are often oversized with a significant margin to meet plateau production demands, often resulting in suboptimal efficiency, especially during the decline phase when utilization falls well below rated capacity. As Brayton-cycle gas turbines achieve their highest efficiency near maximum load, efficiency losses at part-load operation are significant.

Rising CO_2 taxation, fuel costs, and energy-transition targets have driven interest in alternative power-generation solutions. In some regions, options such as electrification from shore, carbon capture and storage (CCS), and the integration of renewable energy sources (e.g., offshore wind) have been explored to reduce upstream CO_2 emissions. Combined-cycle gas turbines and, in some cases, Rankine-cycle systems have also been proposed to improve energy recovery through waste-heat utilization. However, many of these solutions require significant infrastructure upgrades, increase system complexity, or introduce new production-critical elements, limiting their applicability across the global offshore asset base.

In contrast, the large amount of energy dissipated during wellstream pressure let-down remains an underutilized opportunity for recovery. Subsea and topside choke valves reduce high-pressure wellstream fluids to process pressure, often across multiple stages, yet the potential to harness this pressure differential for useful work is rarely exploited. Capturing even a fraction of this energy could offset fuel-gas consumption, reduce CO_2 emissions, and improve process efficiency without introducing major design- or operational risks.

This study investigates a compact TXG system installed in parallel with a CCV, offering an alternative pathway to convert pressure energy into electrical power through near-isentropic expansion. Beyond power generation, the expansion process provides significant gas cooling, enabling improved condensate recovery and reduced process-cooling requirements. The system can be configured to accommodate field tie-ins and late-life operating conditions, making it applicable across the full production lifecycle.

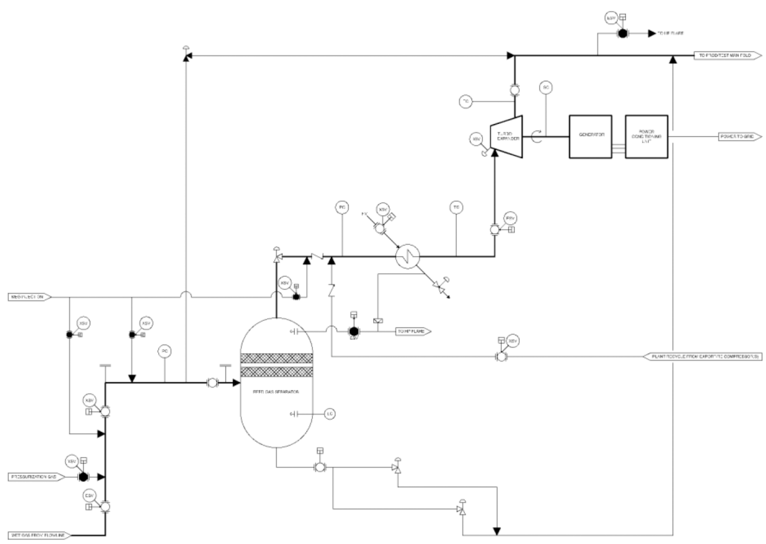

Figure 1—A simplified Process Flow Diagram showing the TXG system during plateau production - thicker lines indicates flow path

Building upon earlier conceptual work, this paper evaluates the technical and economic feasibility of implementing such systems. Using steady-state process simulations over a 23-year production profile, the study quantifies their impact on fuel-gas consumption, CO_2 emissions, condensate recovery, and net economic performance, demonstrating how the TXG system can complement broader decarbonization strategies for offshore gas production.

Methodology

Two steady-state process simulation models were developed: one representing a conventional CCV-based offshore gas processing facility and the other incorporating a TXG system. The objective was to isolate and quantify the impact of the TXG system on energy recovery, fuel gas consumption, process efficiency, and overall plant performance, enabling a clear delta-case evaluation.

Simulation Approach

Both configurations were modeled in an industry-standard steady-state process simulator using identical inlet conditions, plant layouts, and product-quality constraints. The only discrepancy between the two cases was the pressure-reduction device: a conventional choke valve in the CCV case versus a turboexpander–generator in the TXG case. A 23-year production profile was simulated using a case-study function, with 24 annual cases representing field conditions from plateau through decline. Simulation outputs included among others fuel-gas consumption, power demand, process temperatures, condensate recovery, and gas export volumes. These results were exported for post-processing and economic evaluation.

Reservoir and Flowline Representation

A simplified reservoir and subsea-flowline model were used to calculate wellstream arrival pressures as the reservoir depleted. The model applied approximate correlations to estimate pressure losses through the reservoir, wellbore, and pipeline as a function of depletion and flow rate. This simplified approach ensured consistency between the two processing configurations while avoiding unnecessary complexity, so that performance differences could be attributed solely to the integration of the TXG system.

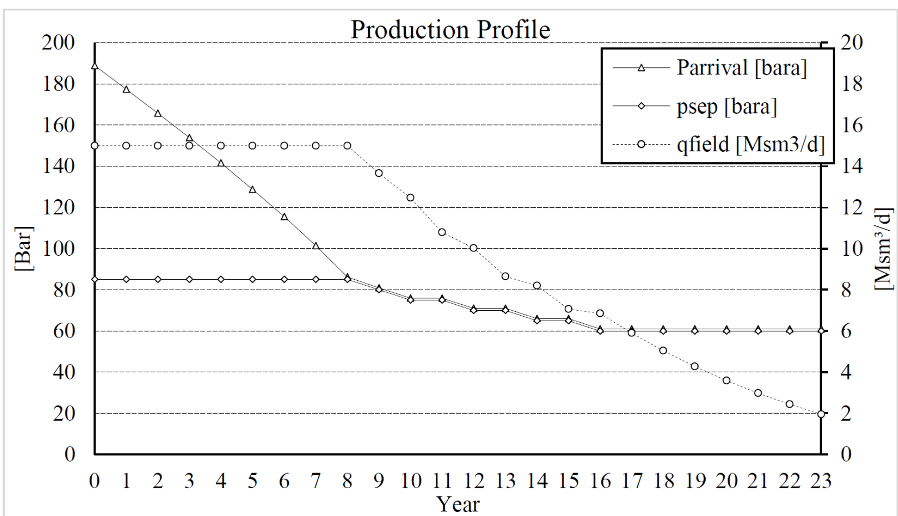

Figure 2—Plot of arrival pressure, inlet separator pressure, and gas flow rate over time

Process Configuration and Constraints

Both process models included three-stage separation, gas dehydration, condensate stabilization, gas compression, and export systems. Frictional losses were accounted for by inserting pipe spools between major process units, and fixed pressure drops were applied across equipment such as coolers, heaters, and separators. These assumptions were based on typical offshore plant designs and were applied uniformly to both models.

Product-quality specifications were applied to replicate real-world requirements. For gas export, a hydrocarbon-dew-point (HCDP) limit was imposed to ensure dense-phase transport without liquid dropout or hydrate risk. Stabilized condensate was constrained to meet a target vapor-pressure limit suitable for safe storage and transport. The simulation models were configured to converge only when these product specifications were satisfied. An arbitrary gas composition representative of a typical North Sea gas-condensate field was applied to both cases.

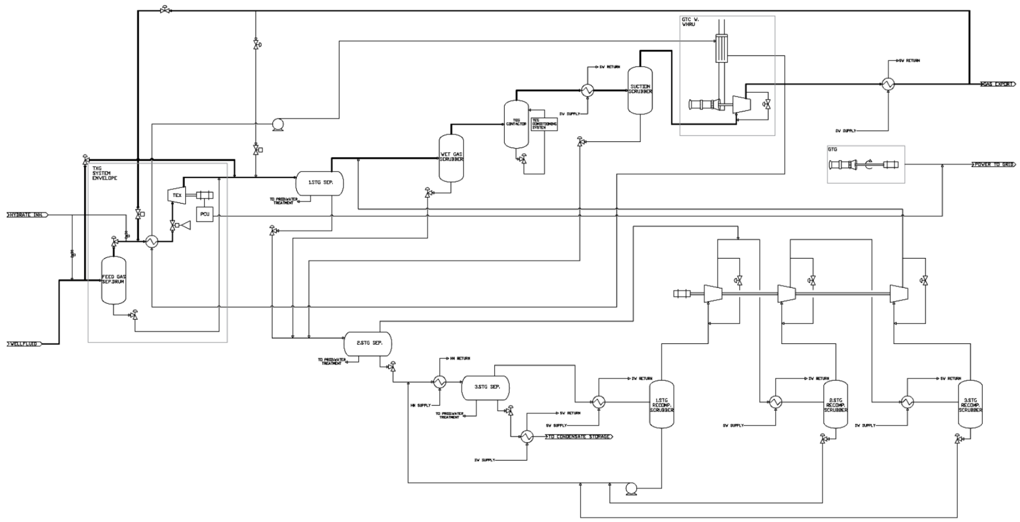

Figure 3—Simplified process flow diagram illustrating the main process equipment used in the simulation models

Power and Fuel-Gas Modeling

Total power demand for the facilities was modeled as the sum of a fixed base load and a variable component associated with seawater lifting and cooling. The base load, representing utilities such as pumps, lighting, and auxiliary systems, was assumed constant throughout the field life. In contrast, seawater pump power demand, operated with variable-frequency drives, was modeled to vary with production rates and cooling requirements, increasing during peak throughput years and decreasing as production declined.

For the TXG case, the power generated by the turboexpander was treated as an offset to the facility's total power demand. This effectively reduced the load on the gas-turbine generator (GTG) that provides power to the plant.

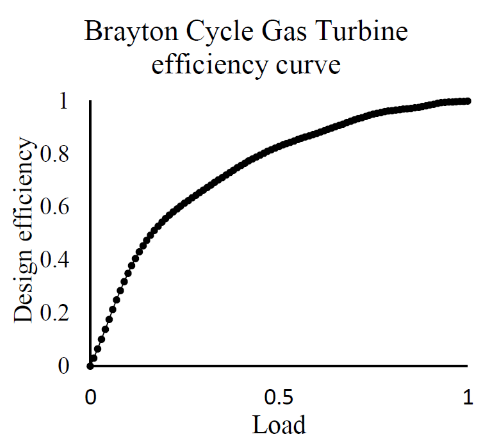

Figure 4—Modeled gas turbine efficiency as a function of load factor

The Gas-turbine driver for the export compressor and GTG were modeled as industry standard Brayton-cycle units. Their full-load capacities and efficiencies were based on representative gas-turbine models widely used in the industry, and part-load performance was calculated dynamically within the simulation software. This approach provided a reasonable approximation of fuel gas consumption across varying operating conditions.

Below a table is with the turbomachinery used in the simulation scenarios.

Table 1—List of turbomachinery used in the simulation scenarios

Brayton cycle turbine	ISO-rated max performance [MW]	Rated efficiency at full load
GTG	8,18	34%
GTC	25,06	37%

Economic Evaluation

To quantify the economic benefit of the TXG system relative to the CCV configuration, a NPV analysis was performed over the 23-year production period. The analysis evaluated incremental cash flows between the two configurations, including:

Fuel-gas savings resulting from reduced gas-turbine operation.

Additional condensate recovery is attributed to the lower separator temperatures enabled by the TXG system.

Avoided CO_2 tax costs, where applicable, based on reduced fuel-gas consumption and associated emissions.

Modeling Approach

To address uncertainty in future commodity and emission prices, a Monte Carlo simulation with 3,000 iterations was performed. Each iteration sampled from predefined probability distributions for natural gas, condensate, and CO_2 emission prices, generating a range of NPV outcomes. The resulting distribution was analyzed to determine key percentiles (P10, P50, P90) and to evaluate sensitivities to input parameters.

Economic Assumptions

Table 2—Economic Inputs

All assumptions were converted to USD using a fixed exchange rate of 7.5 NOK/USD (Table 2).		
Parameter	Value / Range	Notes
Natural gas price	0.20 - 0.93 USD/Sm^3	Uniform distribution
Condensate price	20 - 120 USD/bbl	Uniform distribution
CO_2 emission price	133 - 373 USD/t	Uniform distribution (EU ETS + Norwegian CO_2 Fee)
NOx emission price	3,200 USD/t	Fixed
Discount rate	8 %	Nominal
Prime equipment cost (TXG)	17.3 MUSD	Turboexpander-generator units
Installation & commissioning	22.7 MUSD	Offshore installation & commissioning
Technology development	26.7 MUSD	Qualification, testing, maturation
Power-export infrastructure	13.3 MUSD	Export cable & auxiliaries (optional)
Annual OPEX	0.52 MUSD	3 % of prime equipment cost
Compressor re-bundle (CCV)	6.7 MUSD	One-time (year 13)
TXG re-bundle (plateau)	1.2 MUSD	Years 3 & 6
TXG re-bundle (decline)	0.67 MUSD	Years 10, 16 & 18

Abatement Cost

An equivalent CO_2 abatement cost was derived by setting the CO_2 price/fee to zero and then dividing the resulting NPV by abated CO_2 emissions. A negative abatement cost indicates that the project is profitable without considering CO_2 pricing, whereas a positive cost represents the break-even quota price at which the project becomes economically neutral.

Results

Base-Case Conventional Choke Valve Operation

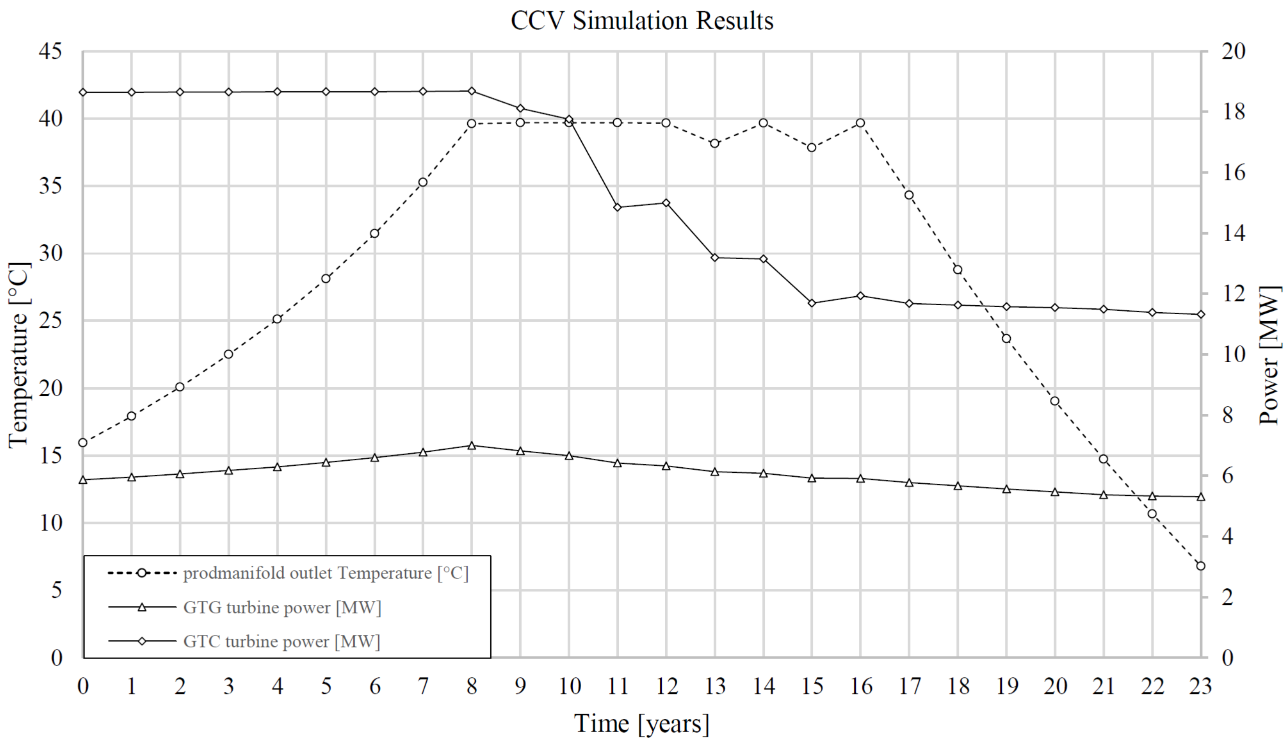

Figure 5—Key output parameters from the CCV case simulation

The CCV configuration served as the reference case for evaluating the performance of the TXG system. In this arrangement, wellstream pressure was reduced isenthalpically, dissipating the available energy without performing useful work.

During the plateau phase (years 0-8), a Joule-Thomson cooling effect was present at the inlet separator, gradually diminishing as reservoir pressure declined toward the end of plateau production in year 8. In the subsequent years, separator-inlet temperatures stabilized at approximately 40 °C, increasing the demand for process cooling.

By year 13, it became economically justified to re-bundle the export compressor to avoid continuous gas recycle that would otherwise have been required to maintain the compressor within its operational envelope. While this intervention reduced the need for sustained recycling, it caused the compressor's gas-turbine driver to operate well below its design load for the remainder of the field life. Prolonged part-load operation reduced turbine efficiency, thereby increasing operating costs and associated CO_2 emissions.

From year 17 onward, gas recycle from the export-compressor discharge to the inlet separator was introduced to maintain the minimum gas velocity through the plant, ensuring adequate liquid entrainment. This recycle flow also provided a modest cooling effect at the inlet separator due to pressure-reduction expansion, an effect that increased as recycled gas gradually dominated the flow regime with declining wellstream production toward the end of the field life.

Peak power demand occurred near the end of plateau production, driven by high throughput and the absence of expansion-induced cooling. Beyond this point, power demand declined in line with production, though the gas-turbine drivers operated increasingly in inefficient part-load conditions. Overall, the CCV configuration demonstrated the combined challenges of energy dissipation, increased process-cooling requirements, and reduced turbine efficiency during late-life operations.

Turboexpander-Generator Operation

As shown in Figure 6, the near-isentropic expansion of high-pressure wellstream fluids through the turboexpander provided two key process benefits: power recovery and gas-stream refrigeration. The latter resulted in significant lower inlet-separator temperatures compared with the CCV case, reducing the need for additional process cooling and enabling greater condensate recovery.

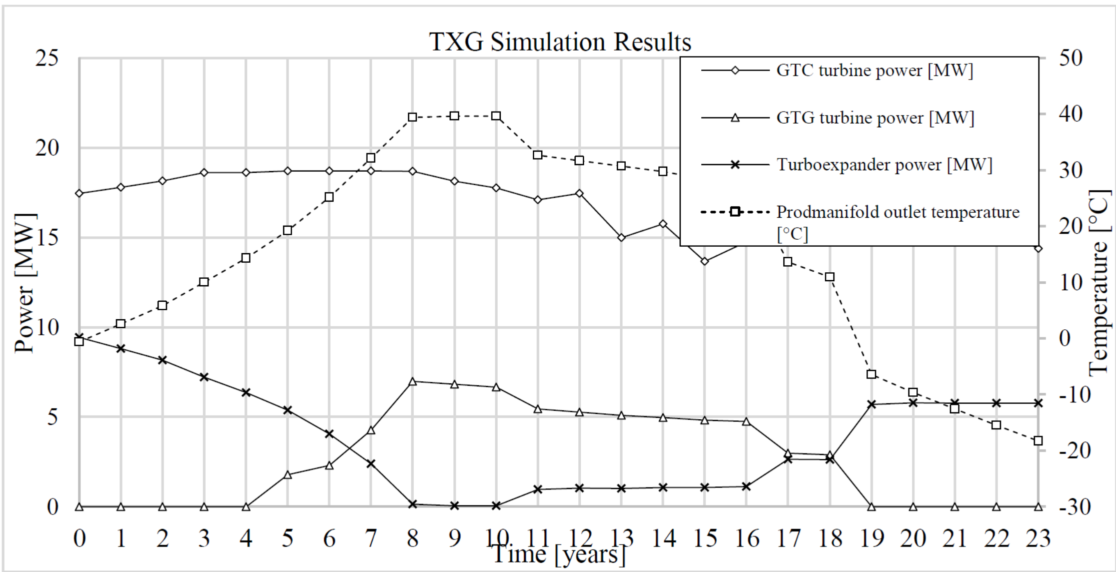

Figure 6—Key output parameters from the TXG case simulation

Turboexpander power production declined gradually during plateau operation (years 0-8) as reservoir pressure dropped. From year 11 onward, a key operational shift was implemented: routing the gas-recycle stream through the turboexpander. As shown in Figure 6, this strategy maintained the export Gas Turbine Compressor (GTC) at high relative load, ensuring acceptable efficiency throughout the field life. Staged bundle replacements ensured that turboexpander performance remained near its design-point efficiency of 88 %, despite changes in flow and pressure conditions.

The net impact on the plant's power demand was significant. TXG generated electricity allowed the GTG to remain offline for approximately 10 of the 23 production years, leading to significant reductions in fuel-gas consumption. From year 19 onward, the GTC effectively served as an indirect power-generation unit, operating at high relative load and maintaining acceptable efficiency. This configuration avoided the simultaneous low-load operation of both the GTG and GTC observed in the CCV case during the decline phase, As observed from figure 7.

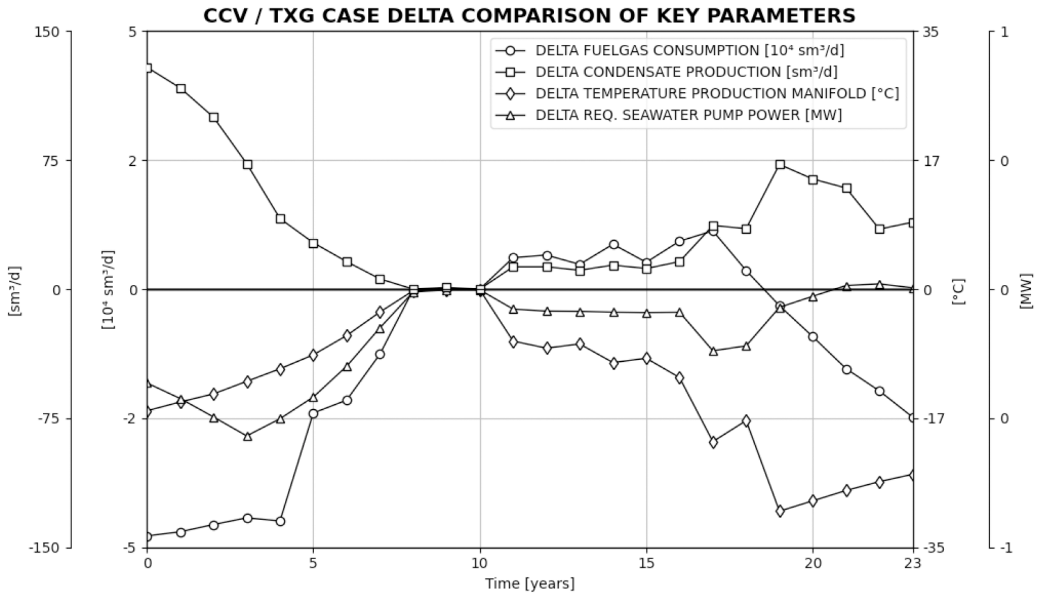

Figure 7—Comparison plot of key delta parameters between the CCV and TXG cases

Economic results

The economic performance of the TXG system delta case was assessed using a Monte Carlo-based NPV analysis over the 23-year production period. Figure 8 illustrates the P10, P50, and P90 percentile NPV development profiles. Across all cases, the breakeven point occurred within 1.5-3.5 years, with substantial value creation during the plateau phase (years 0-8). From years 8-11, the NPV trend remained relatively flat as both facilities operated in a similar manner with the turboexpanders in redundancy until plant recycle was introduced in year 11. From year 19 onward, power surplus again reduced GTG operation, and the enhanced condensate recovery drove a second phase of value creation toward field life end.

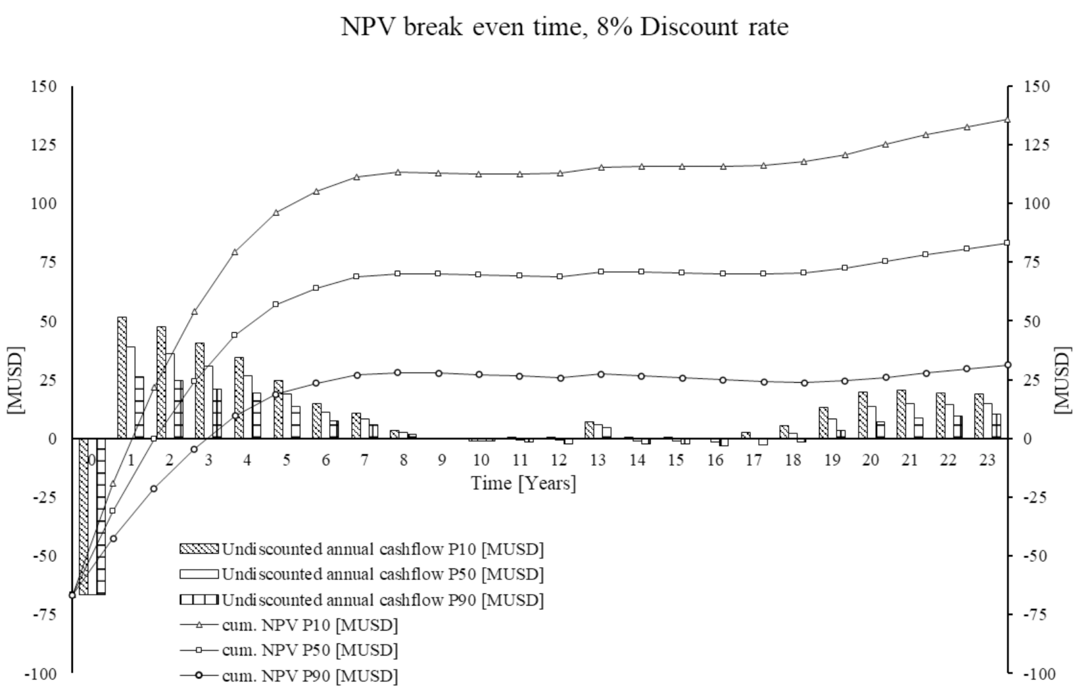

Figure 8—NPV plot of cumulative NPV for P10, P50 and P90 case as a function of time

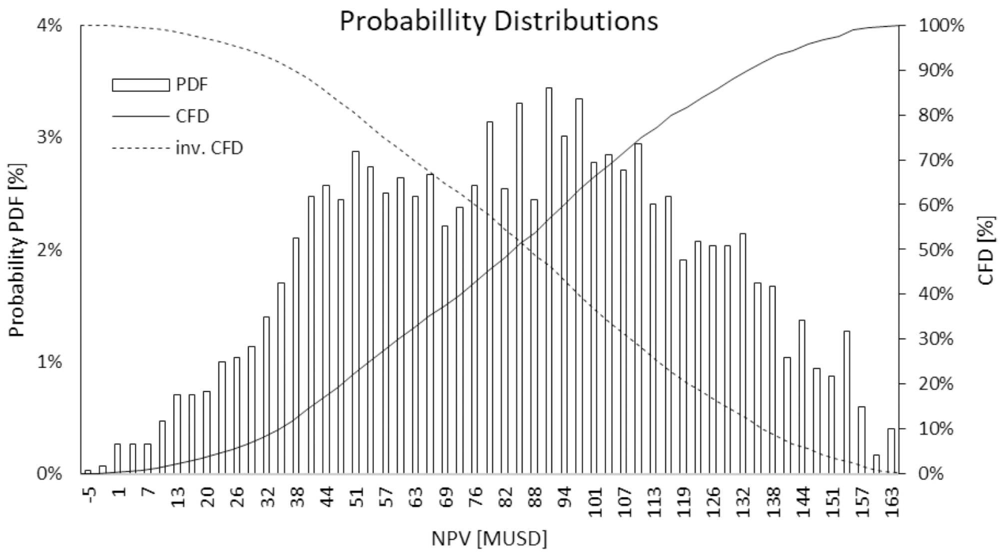

Figure 9—Probability distribution of simulated NPV outcomes, showing PDF (bars), CDF (solid line), and inverse CDF (dashed line)

Table 2 summarizes the key economic indicators. The median (P50) case yielded an NPV of approximately 84 MUSD and an internal rate of return (IRR) of 43 %, with a very low probability (0.26 %) of a negative NPV outcome. The CO_2 abatement cost was negative in the P10 and P50 cases, indicating that the project is profitable even without considering carbon-pricing incentives.

Summary of Economic Results

Table 3—Summary of economic results

Metric	P10	P50	P90
NPV **(MUSD)**	~132	~84	~36
Internal rate of return	63 %	43 %	23 %
CO_2 abatement cost **(USD/t)**	-567	-273	24
Breakeven period	~1.5 yrs	~2 yrs	~3.3 yrs

Discussion

Economic Considerations

The economic evaluation demonstrated that integrating a TXG as a replacement for CCV is a robust and profitable strategy across a wide range of commodity price and CO_2 fee scenarios. A Monte Carlo analysis with 3,000 iterations indicated a very low probability (≈0.3%) of a negative NPV outcome, underscoring the resilience of the concept even under unfavorable market conditions.

In the P50 case, the TXG configuration achieved a net present value (NPV) of approximately 84 M USD and an internal rate of return (IRR) of 43%, with a breakeven period of roughly two years. Even in the pessimistic P90 case, the NPV remained strongly positive at 36 M USD. These results demonstrate that the system can deliver consistent economic value across a variety of market environments.

Crucially, the project's equivalent CO_2 abatement cost was negative in the P10 and P50 scenarios (−567 USD/t and −273 USD/t, respectively) and only marginally positive in the P90 case (≈24 USD/t). This means that the system is economically viable even without considering carbon cost savings for the P10 and P50 scenario. In other words, CO_2 reduction is achieved as a co-benefit of a profitable investment, making the technology an exceptionally cost-effective decarbonization measure.

Importantly, the simulations were designed to maximize condensate recovery, rather than to directly optimize for CO_2 abatement. Nevertheless, the TXG configuration still achieved an estimated reduction of approximately 222,000 tonnes of CO_2 over the field life. This dual benefit, enhancing liquid revenue while lowering emissions, underscores the system's capability to align economic performance with environmental objectives. As shown in Table 1, the GTG selected for the simulation scenarios has a maximum output capacity of 8.18 MW. This represents a conservative and highly optimized choice, tailored specifically to the current energy consumption profile of the facility. In practice, operators often select a turbomachine with excess capacity to allow for future flexibility, such as anticipated power demand from potential tie-ins or additional power-consuming equipment that may become economically viable at a later stage. Had a larger turbine been selected in the simulation, the net NPV and associated economic indicators would likely have been even more favorable. This is because the turbine would have operated at a lower utilization rate throughout the scenario, resulting in reduced efficiency and increased fuel gas consumption, particularly in the CCV case.

Technical Considerations

Process safety. Deploying a TXG system at upstream conditions requires careful attention to process safety, equipment design, and flow assurance. This study proposes an inherently safe design philosophy, with both the upstream scrubber, heater and the turboexpander casing rated for full shut-in pressure from the connected wells. A feed gas shut-off valve is incorporated as primary barrier to safeguard expander/generator overspeed scenarios. Additionally, the embedded CCV provides a secondary relief path by bypassing gas when the TXG system approaches critical speed or other safety thresholds.

Reliability. The turboexpander itself is a single-stage, passive radial-inflow turbine, a configuration known for its robustness against erosion, tolerance to mist-phase liquid content, and mechanical simplicity. Inlet gas conditioning is managed by a compact upstream knockout drum, which removes free liquids and solids prior to the heater and expander. Because turboexpanders can tolerate relatively high liquid fractions in mist form, the scrubber design can remain significantly more compact than conventional separation vessels. The upstream heater serves multiple purposes: it increases gas enthalpy to maximize energy recovery from expansion and flashes off residual hydrocarbon liquids before the gas enters the expander, further reducing erosion risk.

Operability. From a system integration standpoint, the TXG module is designed as a parallel bypass enhancement rather than a full replacement for the CCV. Retaining the CCV provides operational redundancy and flexibility. For example, in fields prone to slugging or transient flow behavior following shutdowns, operators can initially route flow through the CCV until temperatures stabilize and steady-state conditions are established. Once stable operation is achieved, production can be gradually diverted to the TXG. Should maintenance or operational issues arise, the CCV can resume its role as the primary choking device, ensuring uninterrupted production.

Choking strategies. The inclusion of a TXG system also influences well and topside choking strategies, particularly in the context of subsea tiebacks or satellite well clusters. In conventional operations, throttling is distributed between topside facilities and individual wells; however, the TXG creates a strong incentive to maximize the pressure ratio across the expander by reducing wellhead choking. For instance, fully opening the weakest well in a cluster and adjusting the remaining wells accordingly, while using the TXG as the primary throttling point, can optimize energy recovery. This revised choking strategy necessitates comprehensive flow assurance assessments, including evaluation of slugging risks due to reduced superficial gas velocities in the flowlines. Additionally, while the increased flowing pressure requires a re-evaluation of hydrate equilibrium conditions, the associated risk is partially mitigated by reduced Joule-Thomson cooling in the subsea system.

Operational Considerations and possibilities

A CCV-based plant is constrained by the wellstream arrival temperature, whereas a facility equipped with a TXG allows operators to actively control the inlet temperature within an operationally defined range. This flexibility enables multiple strategies: for instance, minimizing inlet heating to operate the plant at the lowest feasible temperature, or maximizing waste-heat extraction upstream to recuperate and convert additional thermal energy into power. The operational window between these extremes can be significant and may be exploited for a variety of purposes, enhancing separation efficiency, avoiding glycol injection etc.

The TXG system also provides opportunities to optimize operations around plant-specific constraints. For example, when limited by a HCDP specification on export or re-injection gas streams, the facility can be operated at a lower temperature, allowing the HCDP requirement to be met at a relatively higher operating pressure. This reduces the compression work required for export or re-injection compressors and, in certain cases, may represent the most advantageous operating strategy. Since most gas-processing facilities are constrained by a combination of product specifications, pipeline requirements, and compressor operating envelopes, the ability to fine-tune inlet conditions can yield meaningful performance improvements.

One of the more transformative opportunities with a TXG system lies in avoiding the need to re-bundle export or re-injection compressors as production transitions off plateau. Instead, the declining volumes can be recycled through the turboexpander, maintaining relatively high loads on the GTC drivers. Based on typical gas-turbine, compressor, and expander efficiencies, an overall efficiency of approximately 20% can be achieved through this configuration. During late-life operations, when plant operating pressures are typically reduced to maximize recovery while export or re-injection pressures remain relatively unchanged, the resulting high-pressure ratios provide substantial refrigeration effects. This enables additional heat to be introduced upstream (e.g., via waste-heat recovery), further increasing gas enthalpy and improving the overall utilization of fuel gas. In practice, integrated efficiencies approaching 30% may be achieved through this approach during the decline phase, a compelling improvement compared to operating a standalone gas-turbine generator at low load.

This strategy eliminates the need for costly compressor re-bundling while retaining the full capacity of the compression system, thereby preserving the facility's capability to serve as a host for future tie-ins.

Furthermore, the TXG system is inherently adaptable for future tie-in field developments when the original host field enters its decline phase. This can be achieved by routing production from new tie-in wells through the turboexpander, either as a standalone stream or in combination with recycled gas from the export and recompression systems if the incremental tie-in production alone is insufficient to prevent compressor surge.

Because the TXG module is bypassed by the original host field, it can be equipped with pre-installed flanges and tie-in points to facilitate future integration with minimal disruption. Depending on the properties of the new tie-in field, modifications to key components such as the expander impeller, inlet heater, and upstream knockout drum may be required; however, the majority of the system—including the core power conditioning unit and electrical interface—can typically be reused without extensive redesign.

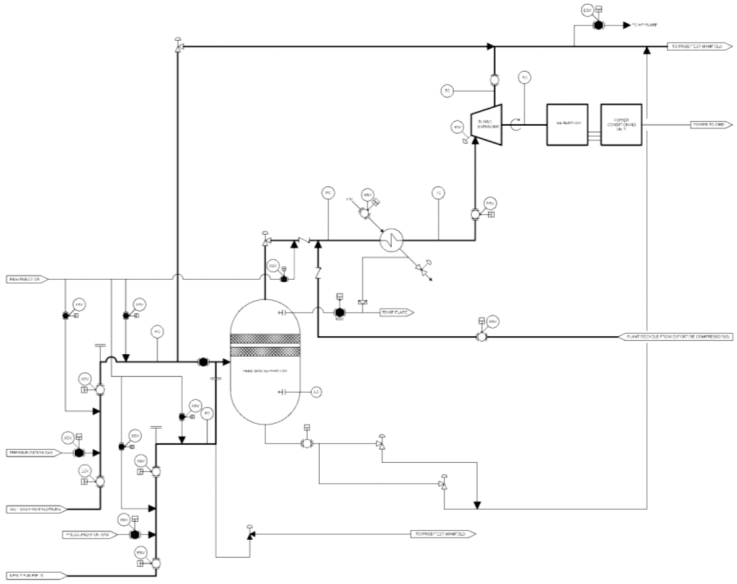

Figure 10—A simplified Process Flow Diagram showing the TXG system with a new tie-in field decline production - thicker lines indicates flow path

In an export compression system, as long as the pressure differential between the export compressor discharge and the new tie-in field is within a suitable range, recycled gas can provide significant increased energy recovery through the expander in combination with the new tie-in field if the additional gas volumes are not sufficient to avoid compressor recycling. This inherent modularity and ability to repurpose existing equipment make the TXG system a valuable enabler for operators, enhancing the attractiveness of the host facility for future subsea tie-ins and satellite field developments.

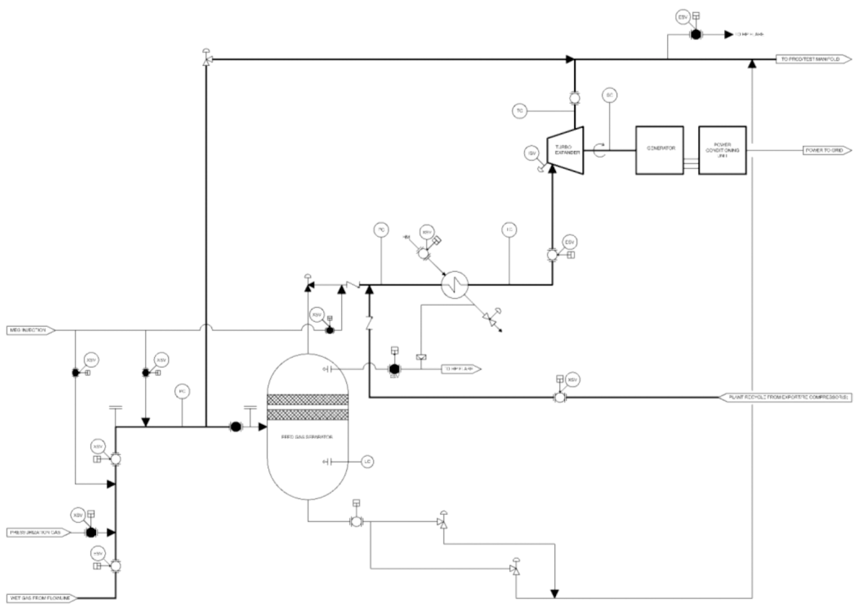

Figure 11—A simplified Process Flow Diagram showing the TXG system in decline production - thicker lines indicates flow path

In another implementation of the system, particularly suitable for single-train facilities comprising one GTG and one GTC, the compressor recycle stream can be deliberately increased to enable the TXG system to supply the full power demand of the process plant, even when the primary production flow is insufficient. This strategy allows preventive or corrective maintenance to be performed on the GTG without necessitating a production shutdown, thereby enhancing plant availability, minimizing downtime, and improving overall operational flexibility.

Limitations of the Study

The simulation approach used in this study incorporates several simplifications, including fixed pressure drops across equipment, simplified compressor and Turboexpander performance curves. Furthermore, the production and process conditions were modeled on an annual basis, with one iteration per year, which represents a relatively coarse temporal resolution. These simplifications were intentional, as the primary objective of the study was to isolate and evaluate the operational and economic impacts of integrating a TXG system. Importantly, the same assumptions and simplifications were applied consistently to both the CCV and TXG configurations, ensuring a fair basis for comparison.

Uncertainties in future commodity pricing also represent a significant source of variability in the NPV analysis. While the selected price ranges for natural gas, condensate, and CO_2 fees were based on historical data and plausible market scenarios, actual future prices could deviate substantially from these assumptions. Nevertheless, because both plant configurations were benchmarked against the same pricing intervals and distributions, the relative differences in performance remain valid and illustrative of the comparative advantages of the TXG system.

Assumptions regarding capital expenditure and operating expenditure also introduce uncertainty. The costs associated with re-bundling compressors in the CCV case and the installation and operation of a TXG system are inherently project-specific. To address this, the study relied on publicly available data and industry experience to derive conservative, benchmark estimates. While these figures may not reflect the exact costs for all facilities, they provide a reasonable basis for evaluating the relative economic impact of the two configurations.

While these limitations should be considered when interpreting the results, they do not undermine the study's overall findings: the comparative evaluation demonstrates that the TXG system offers substantial operational and economic benefits under a range of plausible market and technical conditions.

Proposals for future work

Dynamic Process Modeling

Future studies should incorporate dynamic modeling to capture transient behavior, assess advanced control strategies, and optimize compressor and turboexpander performance for day-to-day operations. This would provide a more comprehensive understanding of the system's operational flexibility under real-world conditions.

Integration Studies

Investigating synergies between the TXG system and other decarbonization initiatives such as partial or full electrification, carbon capture and storage (CCS), or hybrid power generation systems could provide insights into its role within broader net-zero strategies.

Life Cycle Cost Analysis

Performing a detailed life cycle cost analysis, including comprehensive CAPEX and OPEX benchmarking for both greenfield installations and retrofit applications, would improve economic decision-making for deploying TXG systems across various asset classes.

Pilot Scale Testing

Pilot testing of TXG systems in a controlled facility or live production environment would be valuable to validate model predictions, refine operational control strategies, and assess performance during transient conditions.

Integration into Other Asset Classes

Exploring the application of TXG systems beyond upstream gas processing, for example, in LNG or other mid-stream gas processing facilities could reveal opportunities to combine refrigeration needs with power generation, enhancing overall process efficiency.

Conclusions

The integration of a Turbo Expander Generator system as a wellstream expansion and waste energy extraction device offers:

1. Significant reductions in fuel gas consumption and CO_2 emissions
2. Increased condensate recovery and improved energy efficiency
3. Attractive economic performance with high IRRs and short payback periods
4. Flexibility for production and energy-efficiency optimization, as the system significantly expands the range of operational strategies available to the operator
5. Postponement or potential avoidance of CAPEX and OPEX associated with compressor re-bundling, as well as the related operational constraints for future field developments
6. Reduced field OPEX, extending the economically viable field life and thereby increasing the ultimate recovery factor
7. Increased power generation redundancy, as export and/or recompression compressors function as indirect power producers. In some applications, these machines can meet the asset's entire power demand, allowing corrective or planned maintenance on gas turbine generators without deferring production. This also enables optimization between power-generating and compressor driving gas turbines by improving staging and load distribution across the units

This study supports further development, prototype testing, and deployment of turboexpander-generator systems for offshore gas production facilities.

The solution is well-suited for both greenfield and brownfield applications, supporting emissions reduction targets and field life extension.

Substantial and lasting CO_2 reductions in the offshore E&P sector are most likely to be achieved through solutions that are economically self-sustaining, technologically mature, and integrated into the process architecture without being production-critical. Emission-reduction measures should not be treated as isolated add-ons, but as enhancements that improve overall plant performance, reliability, and economics.

Confidentiality Note

Detailed model parameters are documented in Bjerkmo and Bjerkmo (2022). While the thesis remains under temporary confidentiality due to patent considerations, it will be publicly accessible by 2027. Copies may be provided to qualified readers upon request.

Nomenclature

GTG	Gas Turbine Generator
NPV	Net Present Value
CCS	Carbon Capture and Storage
PCU	Power Conditioning Unit

PID	Proportional-Integral-Derivative
TXG	Turboexpander-Generator
CCV	Conventional Choke Valve
HCDP	Hydrocarbon Dew Point
IRR	Internal Rate of Return
CAPEX	Capital Expenditure
OPEX	Operational Expenditure
MW	Mega Watt

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